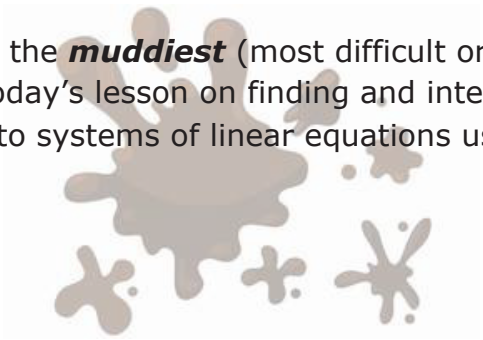




5.7.6 Wrap-Up: Reflection

1. What was the **muddiest** (most difficult or confusing) point in today's lesson on finding and interpreting solutions to systems of linear equations using graphs?



Unit 5, Lesson 8: Solving Systems of Equations by Substitution



Benchmarks

MA.912.AR.9.1 Given a mathematical or real-world context, write and solve a system of two-variable linear equations algebraically or graphically.

MA.912.AR.9.6 Given a real-world context, represent constraints as systems of linear equations or inequalities. Interpret solutions to problems as viable or non-viable options.



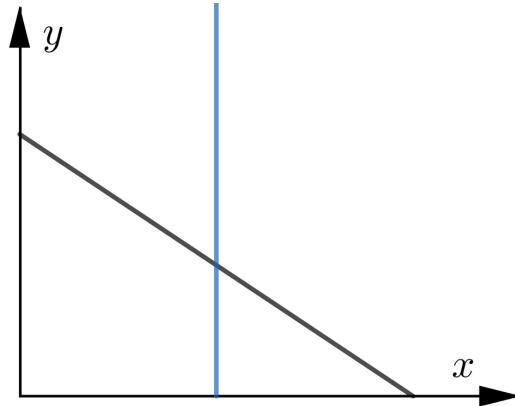
Learning Targets

- ☐ I can solve systems of equations by substitution.
- ☐ I can write a system of equations from a real-world context and solve by substitution.
- ☐ I can interpret the solution to a system of equations in a real-world context.



5.8.1 Warm-Up: Math Talk

1. The graph of a system of linear equations is shown.



For each system given in the table, determine whether or not the system could be represented by the graph and explain how you know.

System	Could the graph represent this system?	Explain
$\begin{cases} x + 2y = 8 \\ x = -5 \end{cases}$	☐ yes ☐ no	
$\begin{cases} y = -7x + 13 \\ y = -1 \end{cases}$	☐ yes ☐ no	
$\begin{cases} 3x = 8 \\ 3x + y = 15 \end{cases}$	☐ yes ☐ no	
$\begin{cases} y = 2x - 7 \\ 4 + y = 12 \end{cases}$	☐ yes ☐ no	



5.8.2 Guided Instruction: Writing and Solving Systems of Equations Using Substitution

While systems of equations can be solved by graphing, they can also be solved algebraically. One method for solving systems of equations algebraically is **substitution**.



The **substitution property of equality** states that if $a = b$, then b may be substituted for a in any expression containing a .

1. Use the system of equations $\begin{cases} y = 2x - 7 \\ x - 3y = 12 \end{cases}$ to complete the following.
 - a. Victor notices from the first equation of the system that the value of y is equal to $2x - 7$, so he substitutes this expression for y in the second equation of the system, as shown.

$$x - 3(2x - 7) = 12$$

Discuss with your partner how Victor's thought process applies the substitution property of equality.
 - b. Solve the equation Victor created using substitution.

$$x - 3(2x - 7) = 12$$

- c. The solution to a system of equations is an ordered pair. Victor's equation was used to find the x -value of the solution. Substitute the x -value into either equation in the system to find the y -value of the solution.

- d. What is the solution to the system $\begin{cases} y = 2x - 7 \\ x - 3y = 12 \end{cases}$?

2. In 1983, the composition of pennies in the United States was changed due, in part, to the rising cost of copper. Pennies minted after 1983 weigh 2.50 grams (g), while the earlier copper pennies weigh 3.11 g. A roll of pennies contains 50 coins. Suppose a particular roll of pennies weighs 132.32 g.

A system of equations can be written to represent this context. To write a system of equations, consider the unknown variables of interest and the information given.

- a. What information is given about the roll of pennies?

- b. Write a two-equation system of equations representing this information, where x represents the number of pennies minted prior to 1983 in the roll and y represents the number of pennies minted after 1983 in the roll.

- c. Solve the system of equations using substitution.

- d. Interpret the solution.



5.8.3 Guided Practice: Writing and Solving Systems of Equations Using Substitution

1. Solve the system of equations using substitution.

$$\begin{cases} x = 3y \\ \frac{4}{3}x - 7y = \frac{9}{2} \end{cases}$$

2. The marketing department at Sea World compared the number of visitors at their Orlando, Florida location with that of their San Diego, California location from 2008 to 2017 to help analyze marketing needs and make predictions.

In 2008, approximately 4.4 million people visited the San Diego location, while approximately 5.9 million people visited the Orlando location. Since then, the number of people visiting each location has been declining — San Diego by 0.1 million people per year and Orlando by 0.2 million people per year.

- a. Name two variables that are being related in this situation.

- b. Write a system of equations to represent the trends at each Sea World Park since 2008.

- c. Solve the system of equations using substitution.

- d. Interpret the solution.



5.8.4 Collaboration: Error Analysis

1. Feng is using substitution to solve the system of equations $\begin{cases} y = 3x - 4 \\ x = -y + 24 \end{cases}$. Feng's partial work is shown.

$$y = 3x - 4$$

$$x = -y + 24$$

$$x = -(3x - 4) + 24$$

$$x = -3x - 4 + 24$$

$$x = -3x + 20$$

$$4x = 20$$

$$x = 5$$

- a. Feng made an error solving for x . Circle the step where he made the error. Then correct his work in solving for x in the system in the space beside his work above.
- b. Using the corrected x -value, find the solution to the system. Show your work.

2. Micah solves the same system, but starts a different way.

Micah's first step is shown. Complete Micah's work to find the solution to the system.

$$y = 3x - 4$$

$$x = -y + 24$$

$$y = 3(-y + 24) - 4$$

3. Discuss with your partner the similarities and differences between Feng's and Micah's solution methods. Then, summarize your discussion in the table.

Similarities	Differences



5.8.5 Your Turn!

1. Cherie receives two different job offers after graduating.
- Job A offers Cherie a salaried position where she will earn \$35,000 total a year.
 - Job B offers Cherie an hourly position where she will earn \$24 per hour as well as an annual bonus of \$500.

The system of equations $\begin{cases} w = 35,000 \\ w = 24h + 500 \end{cases}$ represents this situation, where h is the number of hours Cherie works, and w is her total wages.

- a. Solve the system using substitution.

- b. Interpret the solution.



5.8.6 Wrap-Up: Ticket Out the Door

1. Solve the system of equations using substitution.

$$\begin{cases} 6x - 5y = -8 \\ y = -\frac{1}{2}x + 5 \end{cases}$$

Unit 5, Lesson 9: Solving Systems of Equations by Elimination



Benchmark

MA.912.AR.9.1 Given a mathematical or real-world context, write and solve a system of two-variable linear equations algebraically or graphically.



Learning Target

- ☐ I can solve systems of equations by elimination.